Designing Biopolymer-Bound Regolith Composites for Maximum Compressive Strength

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ABSTRACT

Extraterrestrial construction presents many interesting and new challenges. Transporting large amounts of construction materials from Earth is cost prohibitive. Thus, to take advantage of limited in situ extraterrestrial resources, this work focuses on a novel class of biopolymer-bound soil composite (BSC) materials. These quasi-brittle composites are produced by desiccating a mixture of soil, water, and a biopolymer binder to create a versatile material with uniaxial compressive strength comparable to concrete. This paper expands upon a BSC design methodology for regolith soils. Previous work has been successful in reliably designing BSC materials with graded sands that exhibit compressive strengths above 20 M Pa. Lunar regolith, JSC-1A simulant presents an additional challenge due to the broad ranges of the soil particle sizes. Nineteen BSC mixes using JSC-1A regolith were tested with varying biopolymer solution concentrations and biopolymer to soil ratios. A maximum compressive strength of 25 MPa was obtained at a biopolymer concentration of 48% and a biopolymer to soil ratio of 12.4%. A brief discussion on the effect of manufacturing variables on compressive strength and bulk density is presented. Finally, a design tool is introduced to design regolith BSC for desired compressive strength for future use in the structural design of structures on the Moon, Mars, or other extraterrestial bodies.

1. INTRODUCTION

Over the past 75 years, the consumption of non-fuel materials in the United States has grown from approximately 500 million metric tons to over 4 billion metric tons each year (Matos 2017). The consumption of construction materials, which makes up the majority of non-fuel US material consumption, has grown from approximately 400 million metric tons annually in 1945 to over 3 billion metric tons today. Much of this consumption is in the form of reinforced concrete, a material that serves as the foundation of our built environment and provides the strong backbone of major infrastructure systems.

While reinforced concrete is a critical material for economic development globally, the environmental impacts associated with this material consumption are significant. The manufacture of Ordinary Portland Cement Concrete (OPCC) accounts for 5% of global anthropogenic CO_2 emissions (US EPA 2015).These global warming emissions are in addition to the substantial local and regional environmental impacts associated with the mining of limestone, the calcination of limestone into cement, and the mixing and transportation of concrete to construction sites worldwide (Miller et al. 2016). Hence, there is a need for new construction

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materials that consume less energy to produce, that can be recycled after reaching end of life, and building methods that utilize local and more renewable resources.

Scientific discoveries made in support of space exploration (e.g., LEDs, portable computers, CAT scans) have produced breakthrough technologies that have changed the world (NASA JPL and Satellite Broadband UK 2018).Thus, NASA's current efforts to develop closed-loop building material systems for long-duration missions in the resource-constrained and energy-limited environment of space serve as a useful platform for the development of new construction materials that use local resources, embody ultra-low amounts of energy, and can be continuously recycled.

Inspired by protein-based binders and glues that are common in natural systems and the human body [e.g. (Kamakura 2015; Lee et al. 2007)], Biopolymer-bound Soil Composites, or BSC, are a mixture of granular soil aggregate, biopolymer, and water, which is then desiccated to form a construction material with compressive strengths similar to Ordinary Portland Cement Concrete. By leveraging natural binders and locally sourced aggregates, BSC materials have also demonstrated closed-loop resource consumption, lower energy consumption, and reduced environmental impacts as compared to conventional concrete materials (Roedel et al. 2015). Together, these attributes of suitable mechanical performance and reduced resource and energy use within a circular material system satisfy many of the requirements for both long-duration space missions and highly sustainable terrestrial construction.

This paper translates the BSC design methodology for biopolymer bound sand described by Rosa et al. to lunar regolith using JSC-1A lunar soil simulant as a proxy. In addition to translating the design methodology to lunar regolith, this study seeks to quantify the effect of the manufacturing variables on the mechanical performance of BSC materials. For example, Zeng el a. (2010) have shown a large variability in the density properties of lunar soils as a function of moisture content and compaction method. Thus, in this the effect of the manufacturing variables on the dry bulk density and compressive strength of BSC materials is investigated in order to create an easy-to-use design guide for future experiments and applications of BSC.

2. EXPERIMENTAL DESIGN

As found by Rosa et al. (2019), the maximum compressive strength of BSC materials is a function of the soil maximum bulk density $\rho_{bulk\ soil}$, the maximum concentration in solution of the biopolymer $(\%w/w)_{BP}{}^{max}$, and the biopolymer solution density $\rho_{BP\ Sol.}$. For JSC-1A, the maximum bulk density from literature is 2.01g/cm^3 using the ASTM D4253 with vibration and a surcharge load, and 1.75 g/cm^3 at 13.5% moisture content using the wet compaction method detailed in ASTM 698 (Alshibli and Hasan 2009; Zeng et al. 2010). Additionally, the maximum soil bulk density can be a function of biopolymer solution viscosity, soil type and compaction method (Ayeldeen et al. 2016). Given these parameters, a set of experiments was designed using JSC-1A/Biopolymer BSC mixtures at five biopolymer solution concentrations to (i) determine the mix properties that lead to maximum bulk density of the composite and (ii) translate the BSC mixture theory from sand to JSC-1A soil particles. The BSC mixtures investigated in this study can be found in Table 1.

As proposed by Rosa et al. (2019), BSC mix design begins by determining the biopolymer solution concentration. Dried bovine blood plasma proteins composed primarily of bovine serum albumin (BSA) and immunoglobins were used as the biopolymer, following Rosa et al. The maximum workable biopolymer solution concentration $\%w/w_{BP}$ from previous experiments

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was determined to be approximately 48%, only slightly below the maximum theoretical concentration (taken to be the glass-like transition point) of 55% (Brownsey et al. 2003; Rosa et al. 2020). Hence in this study, mixes with biopolymer solution concentrations of 30%, 38%, 43%, 48% and 53% were investigated. Equation 1 is used to determine biopolymer solution concentration $\%\,w/w_{BP}$.

$$(\%w/w)_{BP} = \frac{m_{BP}}{m_{BP} + m_{water}} = \frac{m_{BP}}{m_{BP\ Sol.}} \tag{1}$$

After selecting the biopolymer solution concentration, the biopolymer solution density for each given concentration must be determined. For biopolymer solution density $\rho_{BP\ sol.}$, the partial specific volume of non-conjugated proteins (0.735 cm^3/g) is used to calculate the density of the gravimetric concentration of plasma proteins using Equation 2 (Durchschlag 1989). The density of water at 25°C is assumed to be 1.003 cm^3/g.

$$\rho_{BP\ sol.} = \frac{1}{\%(w/w)_{BP} * 0.735\ cm^3/g + (1 - \%(w/w)_{BP}) * 1.003\ cm^3/g} \tag{2}$$

Following determination of biopolymer solution concentration and biopolymer solution density, a mix design for high compressive strength requires calculation of the necessary amount of soil mass for the desired volume, along with the maximum soil bulk density associated with the manufacturing method. These values are then used to determine the mass of biopolymer solution needed. In this study, the biopolymer to soil mass ratio $\%w_{BP/S}$ was defined *a priori* as seen in Table 1. The maximum $\%w_{BP/S}$ depended on the estimated saturation ratio of the composite, $V_{BP\ Sol.}/V_v^{min}$, where $V_{BP\ Sol.}$ is the volume of biopolymer solution and V_v^{min} is the minimum volume of voids which occurs at the maximum bulk density. Rosa et al. (2019) found that maximum compressive strength in BSC materials occurs with saturation ratio value of of unity, since oversaturation results in a decrease in bulk density and an increase in shrinkage microcracking and damage. Thus, no mix designs with a $V_{BP\ Sol.}/V_v^{min}$ greater than 1.10 were tested in this study. Knowing the biopolymer concentration $(\%w/w)_{BP}$ (from Equation 1), the biopolymer solution density $\rho_{BP\ sol.}$ (from Equation 2) and the biopolymer to soil mass ratio $\%w_{BP/S}$ (set *a priori*) the saturation ratio of the pores in the composite can be calculated with Equation 3.

$$V_{BP\ Sol.}\Big/V_v^{min} = \frac{\%w_{BP/S}}{(\%\,w/w)_{BP}} \frac{\rho_{bulk,soil}^{max}}{(1 - \frac{\rho_{bulk,soil}^{max}}{\rho_{soil}})\rho_{BP\ sol.}} \tag{3}$$

where ρ_{soil} is the density of the soil particles, $\rho_{BP\ sol.}$ is the density of the biopolymer solution calculated in Equation 2 and $\rho_{bulk,soil}^{max}$ is the maximum bulk soil density expected or achieved using the manufacturing method under investigation. For mix designs in this study, the values measured for JSC-1A by previous researchers were used: $\rho_{soil} = 2.875\ g/cm^3$ and

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$\rho_{bulk,soil}^{max} = 1.75\, g/cm^3$ (Zeng et al. 2010). The bulk density was taken as the one achieved using wet compactions method by Zeng et al. (2010).

Table 1 shows mix designs investigated in this study. The naming convention for the mixes is BPS-X_BS-Y, where X is the biopolymer solution concentration, $\%w/w_{BP}$, and Y is the biopolymer to soil mass ratio, $\%w_{BP/S}$.. Although some mix designs had an estimated saturation ratio greater than 1.10, these mixes were too wet be manufactured with the method described in Section 3 and were discarded**.**

Table 1 – Experimental Mix Design for JSC-1A/BSC Materials

Mix Name	Mix Components (% mass)			$\%(w/w)$	$\%w_{BS/P}$	$\frac{V_{BP\ Sol.}}{V_v^{min}}$	Comment
	$\%w_{Soil}$	$\%w_{Soil}$	$\%w_{Soil}$				
BPS-30_BS-04	88.3%	3.4%	8.3%	29%	3.9%	0.55	
BPS-30_BS-06	84.9%	4.9%	10.2%	32%	5.8%	0.73	
BPS-30_BS-08	79.0%	6.3%	14.6%	30%	8.0%	1.09	Unmanufacturable
BPS-38_BS-04	90.9%	3.5%	5.6%	38%	3.9%	0.40	
BPS-38_BS-06	87.1%	5.0%	7.9%	39%	5.7%	0.60	
BPS-38_BS-08	83.4%	6.3%	10.2%	38%	7.6%	0.80	
BPS-38_BS-10	80.0%	7.6%	12.4%	38%	9.5%	1.01	
BPS-38_BS-11	75.6%	9.8%	14.6%	40%	13.0%	1.30	Unmanufacturable
BPS-43_BS-04	91.9%	3.5%	4.6%	43%	3.8%	0.35	
BPS-43_BS-06	88.4%	5.0%	6.6%	43%	5.7%	0.52	
BPS-43_BS-08	85.0%	6.5%	8.6%	43%	7.6%	0.70	
BPS-43_BS-10	81.9%	7.8%	10.3%	43%	9.5%	0.88	
BPS-43_BS-11	79.0%	9.0%	12.0%	43%	11.4%	1.06	
BPS-43_BS-04	92.6%	3.6%	3.8%	49%	3.9%	0.31	
BPS-48_BS-06	89.4%	5.1%	5.5%	48%	5.7%	0.46	
BPS-48_BS-08	86.3%	6.6%	7.2%	48%	7.6%	0.63	
BPS-48_BS-10	83.5%	7.9%	8.6%	48%	9.5%	0.77	
BPS-48_BS-12	79.4%	9.8%	10.8%	48%	12.3%	1.02	
BPS-48_BS-13	78.2%	10.4%	11.4%	48%	13.3%	1.09	
BPS-53_BS-12	81.0%	10.0%	9.0%	53%	12.3%	0.90	
BPS-53_BS-13	79.8%	10.6%	9.6%	53%	13.3%	0.98	

3. MATERIALS AND EXPERIMENTAL METHODS

3.1 Materials

JSC-1A (major crystalline silicate phases of plagioclase, pyroxene, and olivine with minor oxide phases of ilmenite and chromite and traces of clay) was used as the inorganic aggregate for this experimental study (Zeng et al. 2010). This material was received from NASA Ames Research Center (Moffett Field, CA) and originally manufactured by Orbital Technologies Corporation, Madison, Wisconsin USA . Nineteen independent measurements of the JSC-1A's maximum dry bulk density $\rho^{max}{}_{dry,bulk\ soil}$ were made following a modified ASTM Standard C29 protocol (ASTM International 2017). JSC-1A samples were compacted in two lifts under a

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compression load of 5MPa. The maximum bulk density of the JSC-1A under these laboratory conditions was 1.74g/cm^3 ± 0.01g/ cm^3 which is in agreement with values reported by Zeng et al. (2010).

Bovine blood plasma proteins used in the manufacture of BSC specimens are sold under the tradename *AP920 Bovine Plasma*, and purchased from APC, Inc. (Ankeny, IA). The biopolymer was used as received from the manufacturer. This biopolymer binder was chosen due to its lower environmental footprint and similar mechanical properties when compared to purified *bovine serum albumin* protein studied by others, and also AP920's availability in bulk quantities (Roedel et al. 2015). AP920 contains (by mass) 78% protein, 10% ash, 8% water, with 4% fiber, lipids, and minerals (APC Inc. 2015). The biopolymer water content was measured in the lab to be approximately 5%. This value was used in subsequent mass ratio calculations. Reverse osmosis purified water with a resistance of 18.2MΩ·cm was added to dry mixtures of aggregate and protein.

3.2 Sample Manufacturing

BSC experimental samples were manufactured using a dry-mixing method, wherein dry, pre-weighed amounts of sand and protein were mixed before adding water using a silicone spatula. A total of six samples were created out of each mixing batch. Mixture and specimen masses were measured using a Fischer Scientific A-250 laboratory balance that was accurate to ±10mg. Finally, the mixtures of biopolymer, sand, and water were blended with a KitchenAid Stand Mixer for between five to ten minutes until homogeneous.

A custom molding apparatus, consisting of a cylindrical mold with a 25.4 mm inner diameter and 76mm height made from 316 stainless steel, was used to form the cylindrical BSC samples. A matching piston with a 25.2 mm diameter and 83 mm height was used to compact the samples. Freshly mixed BSC material was loaded into the cylinder and compacted in two lifts, allowing for a specimen height-to-diameter ratio of 2:1. The compaction regime used compacted the sample under a pressure of 5 MPa, prior to inverting the mold 180° (end over end) and compacting the specimen once again under a pressure of 5 MPa at a rate of 0.2 mm/s (strain rate: 4×10^3 1/s). After compaction, specimens were extruded from the mold, weighed, and placed in a Fisher Scientific IsoTemp Oven at 60°C (R.H. 10%) for an additional 24 hours to completely desiccate.

Once fully desiccated, each specimen's top and bottom surfaces were sanded parallel using a disc sander until achieving a maximum angle with the horizontal of less than 0.5° in accordance to ASTM C29/C39M-18. Specimen diameter and height dimensions were measured using digital calipers accurate to ±0.05mm, taking 8 height measurements at the edges and 6 diameter measurements, taking two perpendicular measurements at the top, middle and bottom of the sample. Dry mass measurements were also taken using a Fischer Scientific A-250 laboratory balance accurate to ±10mg and used in conjunction with the caliper measurements to calculate desiccated specimen density.

3.3 Sample Testing

Unconfined uniaxial compression testing of BSC samples was conducted using an MTS Criterion Model 43 electromechanical universal testing machine (Model 50-292-603) and a 10kN load cell. Digital axial displacement measurements were taken from the extension of the

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testing machine crossbar. Samples were loaded at a rate of 0.02mm/s ($\dot{\varepsilon}4 \times 10^{-4}$ 1/s**)**. Ultimate compressive strength (σ_c) was taken as the peak load divided by the measured average cross-sectional area of the cylindrical specimen prior to testing.

4. EXPERIMENTAL RESULTS

The results of compressive strength tests and density measurements are shown in Table 2 and shown graphically in Figure 1 and Figure 2. The dry bulk density was calculated using Equations 4 using the biopolymer to soil ratio, $\%\boldsymbol{w_{BP/S}}$ and the dry densities of BSC, measured experimentally.

$$\rho_{bulk\ soil}^{dry} = \frac{m_{soil}}{V_{BSC}^{dry}} = \rho_{BSC}^{dry} * \frac{1}{1 + \%w_{BP/S}} \tag{4}$$

Table 2 – JSC-1A/AP920 Material Characteristics and Mechanical Properties

Mix Name	Biopolymer Conc. (% w/w)	Biopolymer Soil Ratio $\%w_{BP/S}$	Moisture Content m_c	Saturation Ratio $\frac{V_{BP\ sol.}}{V_v^{min}}$	Dry Density ρ_{BSC}^{dry} (g/cm^3)	Dry Bulk Soil Density $\rho_{bulk\ soil}^{dry}$ (g/cm^3)	Compressive Strength σ_c (MPa)
BPS-30_BS-04	29%	4%	9.4%	0.55	1.74 ± 0.01	1.68	7.3 ± 0.50
BPS-30_BS-06	32%	6%	12.0%	0.73	1.77 ± 0.02	1.67	12.7 ± 1.23
BPS-38_BS-04	38%	4%	6.2%	0.40	1.72 ± 0.02	1.66	7.6 ± 0.57
BPS-38_BS-06	39%	6%	9.1%	0.60	1.78 ± 0.01	1.68	12.5 ± 0.24
BPS-38_BS-08	38%	8%	12.2%	0.80	1.86 ± 0.01	1.73	17.1 ± 0.51
BPS-38_BS-10	38%	10%	15.5%	1.01	1.89 ± 0.02	1.72	17.6 ± 1.06
BPS-43_BS-04	43%	4%	5.0%	0.35	1.74 ± 0.02	1.68	8.8 ± 0.36
BPS-43_BS-06	43%	6%	7.5%	0.52	1.78 ± 0.01	1.68	13.1 ± 0.26
BPS-43_BS-08	43%	8%	10.1%	0.70	1.83 ± 0.01	1.70	15.5 ± 0.50
BPS-43_BS-10	43%	10%	12.6%	0.88	1.91 ± 0.01	1.74	20.6 ± 0.45
BPS-43_BS-11	43%	11%	15.2%	1.06	1.94 ± 0.02	1.75	19.7 ± 1.21
BPS-48_BS-04	49%	4%	4.1%	0.31	1.71 ± 0.01	1.65	4.1 ± 0.28
BPS-48_BS-06	48%	6%	6.2%	0.46	1.76 ± 0.00	1.67	10.0 ± 0.76
BPS-48_BS-08	48%	8%	8.3%	0.63	1.81 ± 0.00	1.68	14.6 ± 0.48
BPS-48_BS-10	48%	10%	10.3%	0.77	1.86 ± 0.01	1.7	17.3 ± 1.00
BPS-48_BS-12	48%	12%	13.6%	1.02	1.97 ± 0.01	1.75	25.4 ± 0.73
BPS-48_BS-13	48%	13%	14.6%	1.09	1.91 ± 0.02	1.69	17.7 ± 1.21
BPS-53_BS-12	53%	12%	11.1%	0.90	1.86 ± 0.02	1.65	14.7 ± 1.77
BPS-53_BS-13	52%	13%	12.0%	0.98	1.89 ± 0.01	1.67	18.6 ± 1.38

To understand the relationship between mixture variables (i.e., void saturation ratio, moisture content, and biopolymer to soil ratio) and JSC-1A/AP920 BSC material properties, these mixture

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variables are plotted against the composite's dry bulk density (Figures 1(a) through 1(c)) and compressive strength (Figures 1(d) through 1(f)) for mixes with varying biopolymer solution concentration. From Figure 1 it can be observed that mixes with a 53% biopolymer solution concentration do not follow the trend observed in other BSC mixes of higher bulk density and higher biopolymer to soil ratio resulting in higher strength. This confirms what was observed by Rosa et al. (2019) that the theoretical biopolymer solution concentration limit based on the glass-transition point of BSA may not be achievable with manufacturing methods that are currently used and the relatively lower bulk density and compressive strength is due to undissolved biopolymer suspended in the biopolymer solution. Due to this uncertainty, data from BSC mixes with 53% biopolymer concentration will be excluded from the analysis.

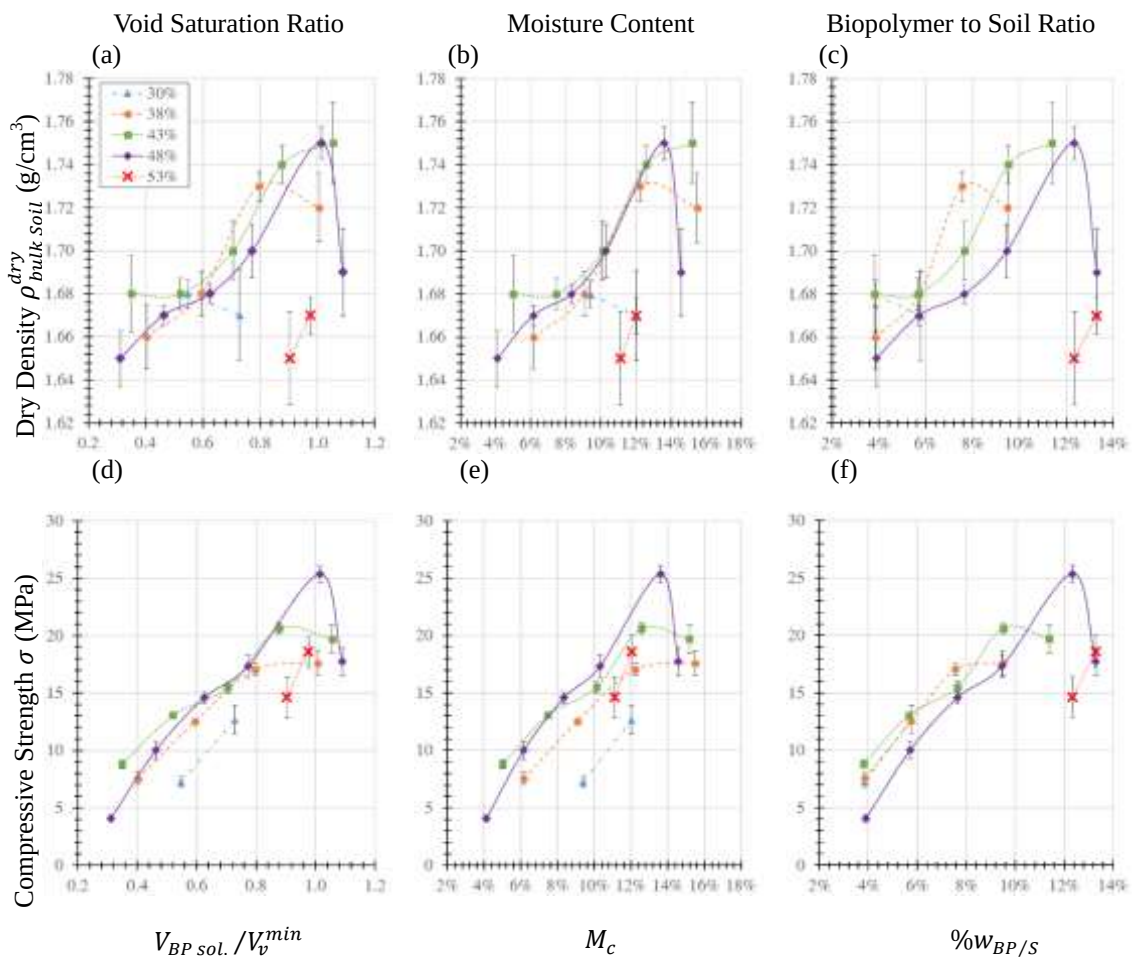

Figure 1 – JSC-1A/AP920 BSC's compressive strength versus (a) void saturation ratio, (b) moisture content, (c) biopolymer to soil ratio, and BSC dry bulk density versus (d) void saturation ratio, (e) moisture content, and (f) biopolymer to soil ratio. The lines connect mixes with the same biopolymer solution concentration as shown in the legend.

From Figures 1 (a) and (b), saturation ratio (proportional to biopolymer solution volume) and moisture content affect bulk density regardless of biopolymer concentration ratio, as shown by overlapping data lines. This is due to the fine particles in JSC-1A interacting electrostatically

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with the biopolymer particles in a process akin to that characteristic of clays, resulting in a pronounced proctor behavior (Chang et al. 2015; Proctor 1933). Additionally, oversaturating the mixture ($V_{BP\ Sol.}/V^{v}_{min} > 1$) generally decreases bulk density. The maximum bulk density observed was 1.75 g/cm^3 at approximately 14% moisture content, near that previously observed in the literature for wet compaction, although the manufacturing method differed (Zeng et al. 2010). From Figure 1 (c), it can be seen that at constant biopolymer to soil ratio, an increase in biopolymer solution decreases bulk density. This is in line with Ayeldeen et al. (2016) who showed that for silt/biopolymer mixtures, an increase in viscosity decreases the bulk density and the biopolymer's viscosity increases with concentration (Bernhardt and Pauly 1980)

Looking at Figure 1 (d) through (f), a pronounced decrease in compressive strength relative to biopolymer to soil concentration also occurs at saturation ratios greater than 1. Hence the data points for over saturated mixes at 48%, 43% and 38% biopolymer concentration are not used to create the mixture design equations. Based on the remaining data of interest, the bulk density and compressive strength of each mix was plotted against the main manufacturing variables, biopolymer solution concentration and biopolymer to soil ratio in Figure 2.

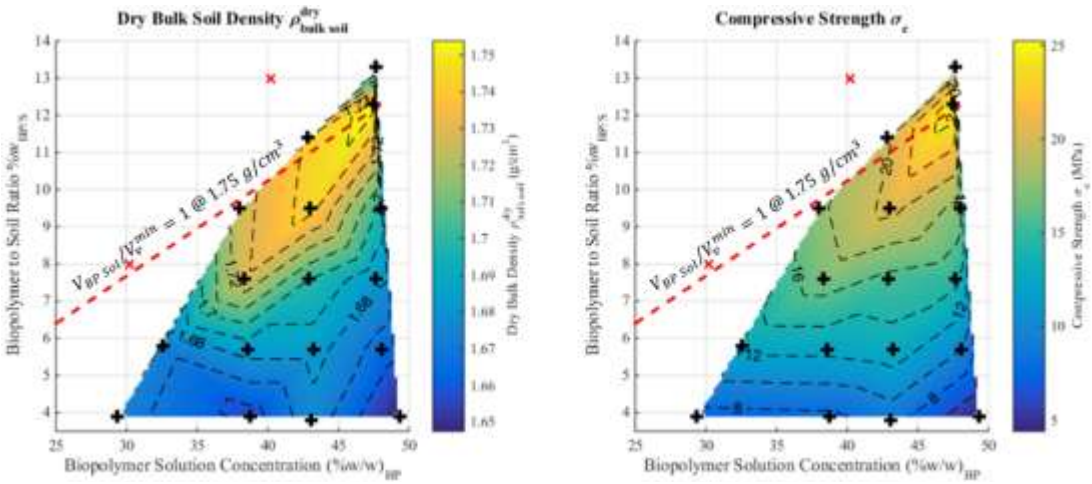

Figure 2 - Interaction of dry bulk density and compressive strength with manufacturing variables for mixes with a biopolymer solution concentration less or equal to 48%. Isolines of the linear interpolation are shown as dashed black lines and the data points as black crosses. The mixes that were unmanufacturable (see Table 1) are presented as red crosses.

5. MODELS FOR PREDICTING BULK DENSITY AND COMPRESSIVE STRENGTH OF JSC-1A BSC MATERIALS

To better understand the relationship between bulk density and biopolymer solution concentration and biopolymer to soil ratio in JSC-1A/AP920 BSC materials, two models where fitted to the experimental data of interest (Figures 3 and 4). The first model is linear while the second is a mixed effects model to capture the interaction between the main variables (Equations 5 and 6).

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$$\rho_{bulk,soil}^{dry} = a * \%w_{BP/S} + b * (\%\, w/w)_{BP} + c$$
$$\text{with } a = 1.12, \quad b = -0.134, \quad \& \ c = 1.67 \tag{5}$$

$$\rho_{bulk,soil}^{dry} = a * \%w_{BP/S} + b * (\%\, w/w)_{BP} + d * (\%\, w/w)_{BP} * \%w_{BP/S} + c$$
$$\text{with } a = 2.67, \quad b = 0.0415, \ d = -3.41 \quad \& \ c = 1.59 \tag{6}$$

The linear and "mixed effects" models for dry bulk density have R-squared values of 0.789 and 0.804, respectively, and their fit is shown visually in Figures 3 and 4. As the R-square values and Figures 3 and 4 illustrate, neither is a perfect with fit but the mixed effect is slightly better Equations 4 and 5 are limited to the prediction of JSC-1A/AP920 BSC bulk density using the manufacturing method described previously in Section 3.2, but application of this methodology can provide insights for future researchers exploring new manufacturing techniques such as extrusion.

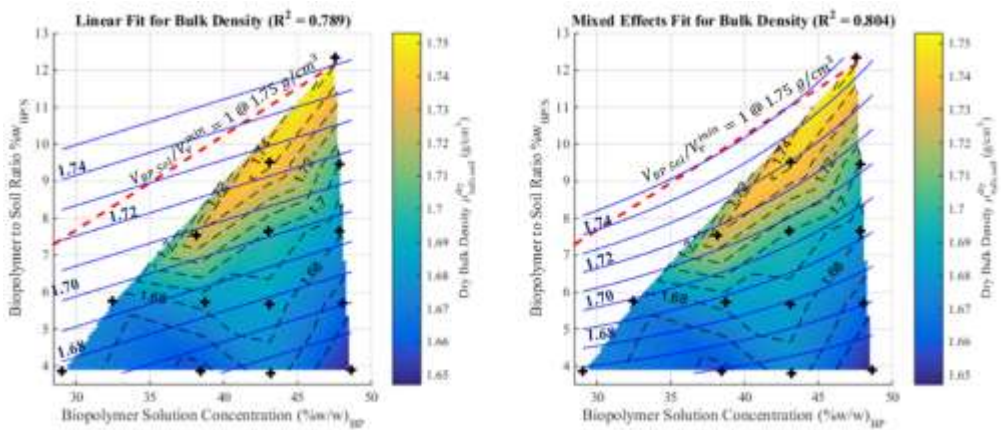

Figure 3 – Linear (left) and mixed effects (right) models predicting dry bulk density to manufacturing variables for compression manufacturing of JSC-1A BSC materials in two lifts at 5 MPa. Isolines of model shown in blue and isolines of interpolation shown as dashed black lines. Data points used to fit the model are presented as black crosses.

A more general model to predict compressive strength of BSC materials manufactured from JSC-1A simulant and dried plasma biopolymer as a function of bulk density and biopolymer to soil ratio was also created. This model is considered to be independent of manufacturing method as shown in Rosa et al (2019). Again, two models, a linear and a mixed effects model were fitted to the experimental data (Equations 7 and 8).

$$\sigma_c = a * \rho_{bulk,soil}^{dry} + b * \%w_{BP/S} + c$$
$$\text{with } a = 62.4, \quad b = 151.2, \quad \& \ c = -102.2 \tag{7}$$

$$\sigma_c = a * \rho_{bulk,soil}^{dry} + b * \%w_{BP/S} + d * \rho_{bulk,soil}^{dry} * \%w_{BP/S} + c$$
$$\text{with } a = 106.7, \quad b = 1139, \ d = -580.5 \quad \& \ c = -177.1 \tag{8}$$

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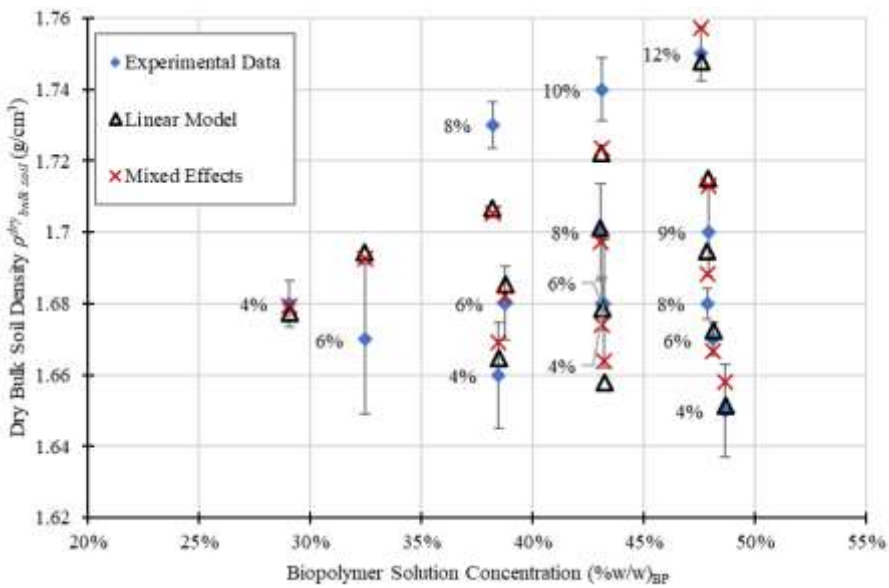

Figure 4 – Comparison of linear and mixed effects models predicting dry bulk density as a function of biopolymer solution concentration. Data labels correspond to the biopolymer to soil ratio of the mixes.

The linear and mixed effects models for predicting JSC-1A/AP920 BSC compressive strength have R-squared values of 0.967 and 0.974, respectively, and their fit is captured visually in Figures 5 and 6. As the R-squared values and Figure 5 and 6 illustrate, both models offer a good prediction of the compressive strength of JSC-1A BSC's. For simplicity, the following discussion will only focus on the linear model.

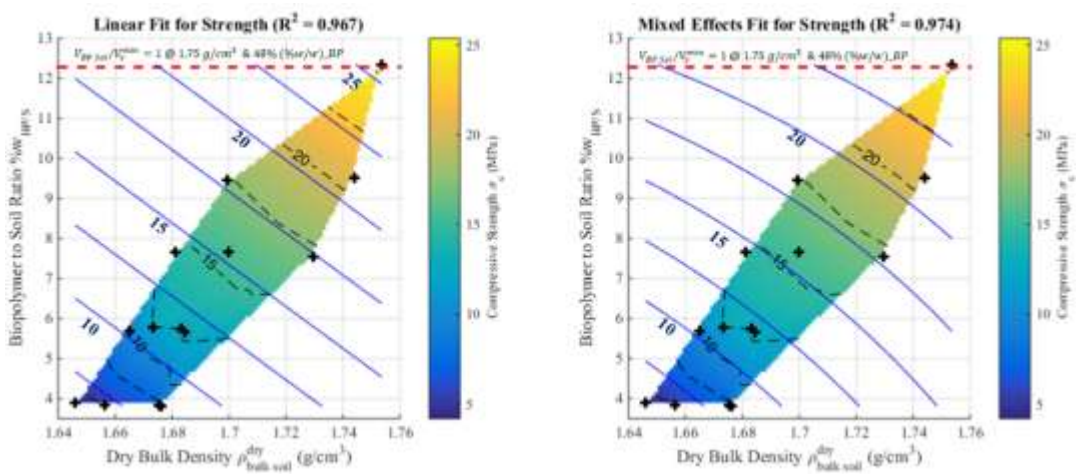

Figure 5 - Comparison of linear (left) and mixed effects (right) models for predicting compressive strength of JSC-1A/AP920 BSC materials as a function of dry bulk density and biopolymer to soil ratio. Isolines of model shown in blue and isolines of interpolation shown as dashed black lines. Data points used to fit the model are presented as black crosses.

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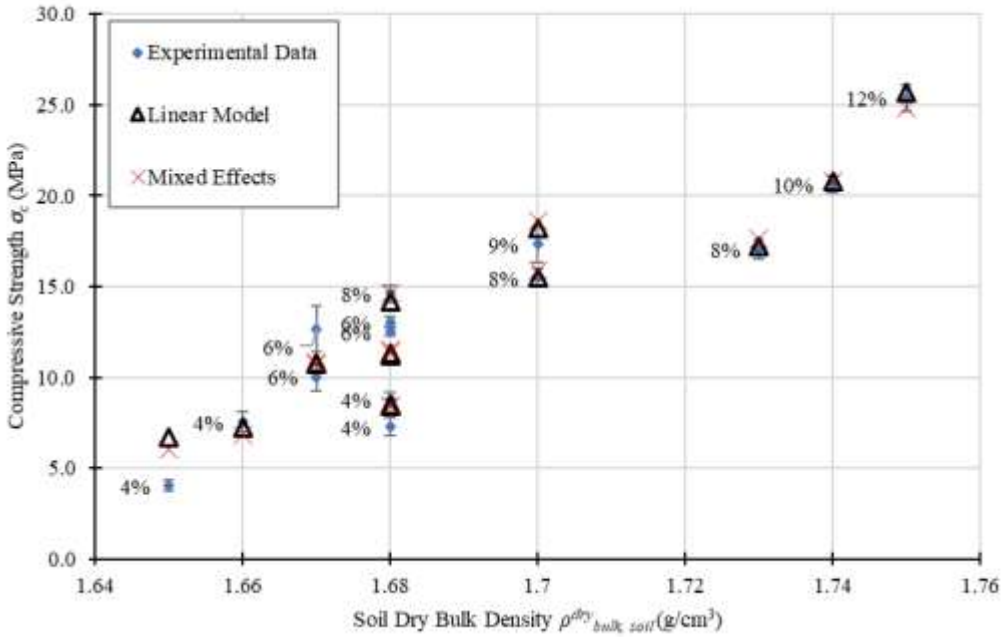

Figure 6 – Comparison of linear and mixed effects models predicting compressive strength as a function of dry bulk density and biopolymer to soil ratio. Data labels correspond to the biopolymer to soil ratio of the mixes.

6. DISCUSSION

As shown in this study, the design theory presented by Rosa et al. (2019) for BSC materials made with sand can be successfully applied to biopolymer-bound soil composites manufactured with JSC-1A, since (i) saturation ratios above unity result in decreased compressive strength, (ii) a linear combination of bulk density and biopolymer to soil ratio can predict their compressive strength at saturation ratios less than or equal to unity. Using the linear compressive strength prediction model, the decrease in strength caused by the bulk density can be estimated and compared to experimental values. Using this approach, hypotheses proposed by previous work to explain the decrease in strength at saturation ratios greater than unity can be confirmed; that damage to the biopolymer matrix due to high shrinkage causes a decrease in compressive strength (Roedel et al. 2019; Rosa et al. 2020).

As hypothesized, the decrease in bulk density does not fully account for the decrease observed in compressive strength. This can be seen more clearly in Table 3. As seen in Table 3, as the saturation ratio increases, so does the difference between the experimental observations of BSC compressive strength and the predictive compressive strength model, hinting at increasing damage in the biopolymer phase with increasing saturation ratio. While the data is preliminary in nature, it supports the decision to eliminate oversaturated mix design data from the compressive strength prediction model. Further work is underway to study the damage in the biopolymer phase at high saturation ratios and high shrinkage.

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Table 3 – Comparison of experimental compressive strength to the linear model predictions for mixes with a saturation ratio $V_{BP\ sol.}/V_v^{min} > 1$.

Mix	$\%w_{BP/S}$	$\rho_{bulk,soil}^{dry}$	$\frac{V_{BP\ sol.}}{V_v^{min}}$	σ_c^{exp}	σ_c^{model}	Difference
BPS-48_BS-13	13.3%	1.69	1.09	17.7 ± 1.21	23.4	-32%
BPS-43_BS-11	11.4%	1.75	1.06	19.7 ± 1.21	24.3	-23%
BPS-38_BS-10	9.5%	1.72	1.01	17.6 ± 1.06	19.5	-11%

The linear compressive strength prediction model (Equation 7, Figures 5 and 6) can also be used to create a design guide for JSC-1A/AP920 BSC materials independent of the manufacturing method[1] by prescribing a bulk density and biopolymer to soil ratio to reach a design compressive strength (Figure 7). As Figure 7 illustrates, the model is applied to the zone where the saturation ratio is less than unity, since as illustrated in Tables 1 and 3, high saturation values are inefficient (yellow zone) or unmanufacturable (orange zone). This design guide can be combined with Equation 6 to create a design guide specifically for the manufacturing method described in Section 3.2 (Figure 8). The model's R-squared value is 0.964 which is comparable with the linear strength model R-squared value of 0.967 (Equation 7).

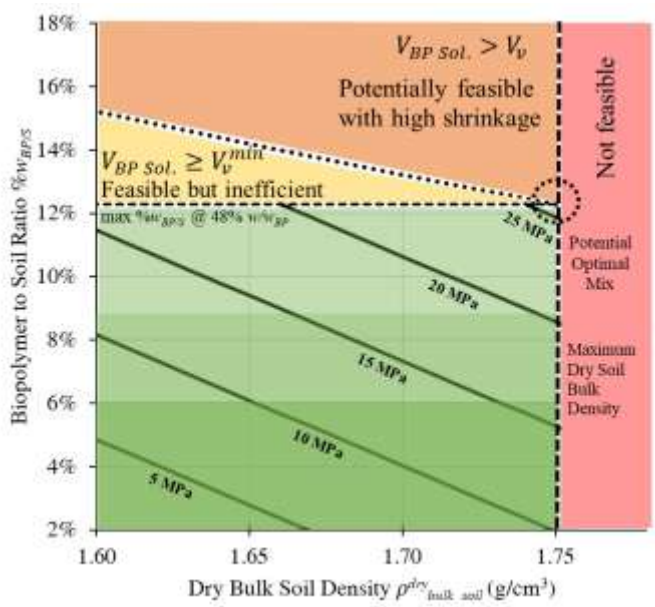

Figure 7 – Design guide for target ultimate compressive strength of BSC materials manufactured using AP920 biopolymer and JSC-1A lunar soil simulant.

[1] Assuming the manufacturing method can produce consistent bulk density throughout the specimen.

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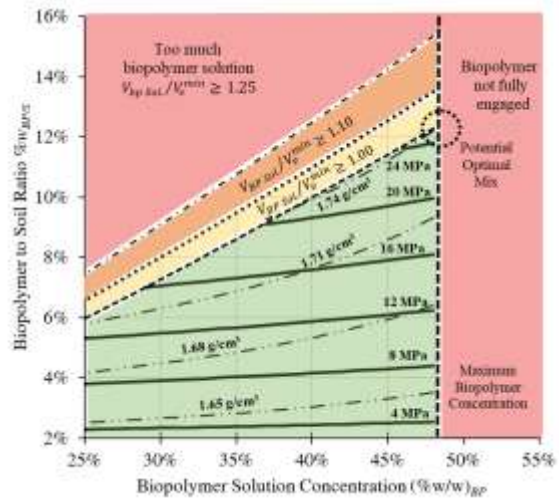

Figure 8 – Combined design guide for achieving targeted ultimate compressive strength of BSC materials manufactured using AP920 biopolymer and JSC-1A lunar soil simulant based on Biopolymer Concentration and Biopolymer to Soil Ratio for double-sided at 5 MPa manufacturing compression (R^2 =0.964)

7. CONCLUSIONS & FUTURE WORK

This work investigated how biopolymer concentration and biopolymer to soil ratio affect both dry bulk density and compressive strength of JSC-1A/AP920 Biopolymer-bound Soil Composites (BSC). Both the solution saturation ratio and the moisture content played a role in the dry bulk density observed the BSC mixes. The resulting density can be predicted from mix design variables using the model presented in this manuscript. Additionally, the maximum bulk density observed corresponds to those previously observed in the literature for wet compaction of JSC-1A lunar soil simulants. Since bulk density depends on the manufacturing method, future work is needed to assess whether alternate methods can bring the bulk density closer to that of dry compaction (1.75 g/cm^3 vs. 2.03 g/cm^3)(Zeng et al. 2010).

JSC-1A/AP920 BSC materials exhibited a linear relationship between compressive strength, dry bulk density and biopolymer to soil at saturation ratio less than or equal to unity, following the theory developed for Sand BSC (Rosa et al. 2020). This discovery enables the creation of design tools for the future deployment of BSC's in both terrestrial and extraterrestrial construction applications where high strength is desired, such as micrometeoroid shielding (Allende et al. 2019) or rural airstrips. Future work will concentrate on characterizing the resulting bulk density of new manufacturing methods, such as 3D printing, to enable the use of this novel material family in a greater range of applications.

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